

Supplementary Material for QMOF-Rec

Representative RAG/LLM Interface Screenshots

The screenshots below illustrate representative query-response behavior in the implemented QMOF-Rec interface. They demonstrate interface behavior, grounded meta-data reporting, missing-descriptor warnings, and refusal of unsupported validation claims. They are not part of the offline ranking validation.

Descriptor Availability and Masking

The reproducible QMOF-Rec evaluation uses 20,372 local vector-metadata records. Descriptor coverage was audited before revision: density is available for 20,372/20,372 records, band gap for 10,810/20,372 records, and void fraction for 0/20,372 records. Void fraction is therefore treated as unavailable rather than as zero, and porosity is excluded from active numerical ranking, diversity distances, and LEA candidate remapping.

Final Diversity-Aware Mask-Aware Rerun Artifacts

The final protocol was rerun with the repository configuration file `configs/final_diversity_aware_masked_protocol.json`. The rerun uses five query scenarios, ten random seeds (42–51), top- $K = 5$, a main candidate-pool size of 100, and candidate-pool sensitivity at 50, 100, and 200. All ranking methods use the same saved candidate pool for a given query and seed. The complete machine-readable outputs are stored under the final diversity-aware artifact directory and include candidate pools, top- K identifiers, aggregate metrics, per-query/seed metrics, query-stratified summaries, variance decomposition, candidate-pool sensitivity, catalog coverage, top- K overlap, list-level comparisons, and the final grounded explanation rerun.

Historical artifacts used in earlier drafts are preserved read-only under the historical artifact directory. The prior final-mask-aware outputs are preserved under the before-diversity-revision directory. The zero-diversity diagnosis is stored in the diversity-revision artifacts and shows that the previous zero-diversity result was caused by binned active utility scores: band gaps between 1.0 and 3.5 eV mapped to the same score, densities between 1.0 and 2.0 g/cm³ mapped to the same score, and the stability proxy often inherited the binned density score. The top CO₂ candidates had different raw values but identical transformed active objective vectors.

The final selected diversity representation is a full-precision hybrid material distance using observed density, observed band gap, formula-derived descriptors, and formula-derived graph proxies. Void fraction remains excluded. In the final rerun, WeightedSum, TOPSIS, MMR, and baseline LEA tie on Rel@5 (0.8479) and NDCG@5 (1.0000). LEA improves diversity relative to WeightedSum (0.0224 versus 0.0187), while MMR is slightly more diverse (0.0241) but much slower. Candidate-pool sizes 50, 100, and 200 give identical aggregate LEA Rel@5 and NDCG@5 in this rerun, with small diversity variation and runtime increasing as pool size increases.

Table 1: Descriptor coverage in the local metadata used for the revised evaluation.

Descriptor	Observed	Total	Coverage
Band gap	10,810	20,372	53.1%
Density	20,372	20,372	100.0%
Void fraction	0	20,372	0.0%

For each material m_i , QMOF-Rec stores an availability mask $\mathbf{a}_i$ in parallel with the normalized objective vector. Weighted relevance is computed only over observed objectives, and distances are computed only over jointly observed dimensions. This prevents a missing descriptor from being interpreted as a physical value of zero.

Retrieval Versus Reranking Missingness

The implemented retrieval layer and the numerical reranking layer use different representations. Retrieval uses SentenceTransformer embeddings of metadata text templates stored in a FAISS L2 index. The retrieval vector is therefore a text embedding rather than the numerical objective vector used by WeightedSum or LEA. In the revised vector-index builder, unavailable fields are written as unavailable metadata rather than as zero. The reproducible offline benchmark tables use a deterministic semantic proxy over metadata text and formula tokens, not a live sentence-transformer retrieval rerun.

Reranking uses active numerical objectives with masks. The active objectives are semantic score, band-gap score, density score, and stability score. Void fraction is excluded from relevance scoring, diversity distances, candidate remapping, objective balance, hypervolume proxy interpretation, and radar plots because it is unavailable for every local metadata record.

Detailed Computational Workflow

Algorithm 1 QMOF-Rec retrieval and objective scoring

Require: Query q , metadata corpus $\mathcal{D}$, QMOF records $\mathcal{M}$, candidate pool size n

Ensure: Candidate pool C_q , objective matrix X_q , weight vector $\mathbf{w}_q$

- 1: Embed query q and retrieve an initial semantic neighborhood from $\mathcal{D}$
 - 2: Construct $C_q = \{m_1, \dots, m_n\}$ from the top retrieved material records
 - 3: Infer or select query-dependent weights $\mathbf{w}_q$
 - 4: **for** each material $m_i \in C_q$ **do**
 - 5: Compute semantic score $s_{\text{sem}}(m_i, q)$
 - 6: Compute available property scores, including band-gap and density scores where descriptors exist
 - 7: Build availability mask $\mathbf{a}_i$; do not impute missing descriptors as numerical zeros
 - 8: Exclude unsupported descriptors from active weighted scoring
 - 9: Form normalized objective vector $\mathbf{x}_i$
 - 10: **end for**
 - 11: Assemble X_q from all $\mathbf{x}_i$ and A_q from all $\mathbf{a}_i$
 - 12: **return** $C_q, X_q, A_q, \mathbf{w}_q$
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Algorithm 2 LEA-guided multi-objective reranking for QMOF-Rec

Require: Query q , candidate pool C_q , objective matrix X_q , availability matrix A_q , weight vector $\mathbf{w}_q$, population size N , iterations T , recommendation length K

Ensure: Top- K recommendation list S_K

- 1: Initialize population P^0 using candidate objective vectors and random vectors in $[0, 1]^d$
 - 2: **for** $t = 1$ to T **do**
 - 3: **for** each individual $\mathbf{z}_i^t \in P^t$ **do**
 - 4: Map to nearest valid candidate $m_i = \pi(\mathbf{z}_i^t)$ using masked distance
 - 5: Compute $f(q, m_i, S)$ using the list-aware utility
 - 6: Generate mutant $\tilde{\mathbf{z}}_i^{t+1}$ using the implemented lotus mutation
 - 7: Map mutant to nearest valid candidate $\tilde{m}_i = \pi(\tilde{\mathbf{z}}_i^{t+1})$ using masked distance
 - 8: **if** $f(q, \tilde{m}_i, S) > f(q, m_i, S)$ **then**
 - 9: Accept $\tilde{\mathbf{z}}_i^{t+1}$
 - 10: **else**
 - 11: Retain $\mathbf{z}_i^t$
 - 12: **end if**
 - 13: **end for**
 - 14: **if** $t \bmod \tau = 0$ **then**
 - 15: Replace weakest ρN individuals by random normalized vectors
 - 16: **end if**
 - 17: **end for**
 - 18: Rank unique mapped QMOFs by LEA fitness and diversity-aware tie-breaking
 - 19: **return** top- K materials S_K
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Algorithm 3 Graph-aware RAG-assisted recommendation workflow

Require: Query q , QMOF graph $\mathcal{G}$, metadata corpus $\mathcal{D}$, material records $\mathcal{M}$

Ensure: Recommended materials with evidence-grounded explanation

- 1: Retrieve candidate pool C_q and evidence set $\mathcal{R}_K(q)$
 - 2: Build or update material graph features $\mathbf{h}_i^{(0)}$
 - 3: Encode QMOF graph with GraphSAGE or GAT when trained models are available
 - 4: Compute optional graph similarity objective $s_{\text{graph}}(m_i, q)$
 - 5: Run objective scoring and LEA-guided reranking to obtain S_K
 - 6: Construct augmented prompt $A(q)$ from evidence, candidate records, scores, and uncertainty flags
 - 7: Generate explanation y using the scientific assistant
 - 8: Record feedback, logs, and analytics for future model improvement
 - 9: **return** S_K, y
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RAG/LLM Scoring Rubric and Hallucination Flags

Table 2: RAG/LLM evaluation dimensions and scoring rubric.

Dimension	Question	Scale	1 (Poor)	3 (Acceptable)	5 (Excellent)
Retrieval relevance	Does the RAG module retrieve relevant QMOF records for the query?	0–5	Irrelevant retrieval	Partially relevant	Directly useful for answering the query
Groundedness	Are the QMOF IDs and claims supported by retrieved candidates?	1–5	No retrieved IDs mentioned; context ignored	About half of IDs overlap with retrieved	All IDs drawn from retrieved candidates
Metadata consistency	Are QMOF IDs, formulas, density, and band-gap values consistent?	1–5	5+ inconsistencies	2 inconsistencies	No inconsistencies
Limitation awareness	Does the answer state when void fraction, porosity, or adsorption data are unavailable?	1–5	Makes unsupported adsorption/porosity claims	No explicit statement, but no false claims	Explicitly and correctly states missing descriptors
Explanation quality	Does the answer explain recommendation rationale?	1–5	Very short, no property discussion	Discusses some properties	Discusses properties, relevance, and uncertainty

Table 3: Binary hallucination flags used in the automated evaluation harness.

Flag	Meaning
Hallucination detected	Any of the flags below is true
Unsupported adsorption claim	Answer states or implies a specific CO ₂ uptake, adsorption capacity, or gas-storage performance value
Unsupported porosity claim	Answer states or implies a specific or measured/confirmed porosity value
Experimental-validation claim	Answer claims a candidate is experimentally or laboratory validated
Graph misrepresentation	Answer describes GraphSAGE/GAT results as CIF-derived atomistic graphs rather than formula-derived metadata graphs
Metadata error	At least one mentioned QMOF ID is not in the retrieved set, or a stated formula does not match retrieved metadata
Missing-descriptor warning	Answer explicitly notes that a requested descriptor, such as void fraction or adsorption data, is unavailable

Repeated-Seed and Candidate-Pool Scope

The repeated-run artifacts use five query scenarios with ten random seeds per scenario. These 50 query/seed rows are useful for robustness checks, but they are nested within five query scenarios and should not be interpreted as 50 independent scientific queries. Candidate-pool-size sweeps are retained as supplementary reproducibility artifacts; they are used to examine retrieval–reranking sensitivity rather than to establish a universally optimal pool size.

Table 4: Query-stratified repeated-seed results for selected methods. Values are means and standard deviations over ten seeds within each query scenario. Seeds quantify stochastic algorithm variation within a fixed query, not independent scientific tasks.

Query	Method	Rel@5	SD	NDCG@5	SD	Div.	SD
q1_co2_adsorption	LEA baseline	0.7599	0.0000	1.0000	0.0000	0.0288	0.0026
q1_co2_adsorption	MMR	0.7599	0.0000	1.0000	0.0000	0.0333	0.0011
q1_co2_adsorption	WeightedSum	0.7599	0.0000	1.0000	0.0000	0.0163	0.0000
q2_photocatalysis	LEA baseline	0.8752	0.0000	1.0000	0.0000	0.0150	0.0000
q2_photocatalysis	MMR	0.8752	0.0000	1.0000	0.0000	0.0150	0.0000
q2_photocatalysis	WeightedSum	0.8752	0.0000	1.0000	0.0000	0.0150	0.0000
q3_lightweight_storage	LEA baseline	0.9174	0.0000	1.0000	0.0000	0.0193	0.0022
q3_lightweight_storage	MMR	0.9174	0.0000	1.0000	0.0000	0.0204	0.0006
q3_lightweight_storage	WeightedSum	0.9174	0.0000	1.0000	0.0000	0.0165	0.0000
q4_balanced_discovery	LEA baseline	0.8979	0.0000	1.0000	0.0000	0.0194	0.0010
q4_balanced_discovery	MMR	0.8979	0.0000	1.0000	0.0000	0.0204	0.0006
q4_balanced_discovery	WeightedSum	0.8979	0.0000	1.0000	0.0000	0.0165	0.0000
q5_insulating_frameworks	LEA baseline	0.7893	0.0000	1.0000	0.0000	0.0294	0.0014
q5_insulating_frameworks	MMR	0.7893	0.0000	1.0000	0.0000	0.0316	0.0018
q5_insulating_frameworks	WeightedSum	0.7893	0.0000	1.0000	0.0000	0.0293	0.0000

Table 5: Direct variance decomposition for mean relevance in repeated-seed artifacts. Between-query variance is much larger than within-query seed variance, supporting the interpretation that query scenario, not random seed, is the main dependence unit.

Method	Between-query var.	Within-query seed var.	Ratio
LEA baseline	0.004815	0.000000	481,544.3
MMR	0.004815	0.000000	481,544.3
Random	0.006060	0.000266	22.8
SemanticOnly	0.006870	0.000071	96.9
WeightedSum	0.004815	0.000000	481,544.3

Table 6: Candidate-pool-size sensitivity for the available repeated-seed LEA pool variants. The saved artifacts contain pool sizes 50, 100, and 200. Pool size 100 is retained as the predefined main experimental setting to maintain consistency with the primary evaluation, not because this sensitivity analysis establishes it as optimal. Pool sizes 50–200 produce broadly similar quality, while runtime increases moderately with pool size.

Pool	Rel@5	NDCG@5	Diversity	Unique top-K frac.	Runtime (ms)
50	0.8479	1.0000	0.0225	1.000	76.94
100	0.8479	1.0000	0.0224	1.000	83.49
200	0.8479	1.0000	0.0229	1.000	98.74

Table 7: List-level check of LEA no-diversity versus WeightedSum in the final diversity-aware rerun. Aggregate Rel@5 and NDCG@5 are identical in all 50 query/seed rows, but top- K lists and ordering are not generally identical.

Check	Rows	Result
Same Rel@5	50/50	yes
Same NDCG@5	50/50	yes
Same diversity	11/50	only where top- K set matches
Same top- K IDs	11/50	exact set match
Same top- K order	0/50	no exact order match

Table 8: Actual catalog coverage in the final diversity-aware rerun. Coverage is the number of unique QMOF records selected across all top- K lists for a method divided by the full catalog size of 20,372 records. This is distinct from Unique@ K , which is 1.000 for every method in the aggregate table.

Method	Unique catalog items	Catalog coverage
WeightedSum	16	0.000785
LEA baseline	32	0.001571
MMR	34	0.001669
ParetoCrowding	50	0.002454
SemanticOnly	163	0.008001
Random	236	0.011585

Table 9: Final local grounded explanation rerun using the final diversity-aware LEA candidate lists. The live OpenAI chat service requires an external API key in this checkout, so the rerun uses deterministic candidate-conditioned explanations with the same no-fabrication constraints and automated rubric.

Scenario	Queries	Ground.	Metadata	Limit.	Halluc.
balanced_discovery	5	5.0	5.0	5.0	0%
co2_adsorption	5	5.0	5.0	5.0	0%
general_recommendation	3	5.0	5.0	5.0	0%
lightweight_gas_storage	5	5.0	5.0	5.0	0%
missing_information_trap	3	5.0	5.0	5.0	0%
photocatalysis	5	5.0	5.0	5.0	0%
wide_band_gap	4	5.0	5.0	5.0	0%
Overall	30	5.0	5.0	5.0	0%

Observed Physical Descriptor Means: All Records Versus Selected Top-K Portfolios

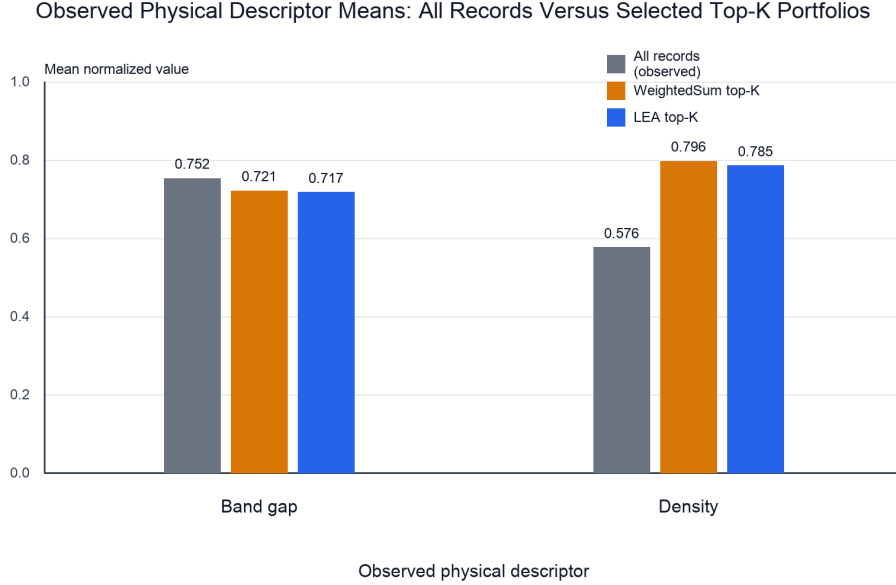


Figure S1: Observed physical descriptor means for all local records versus selected top- K portfolios. The figure includes only band gap and density because these are the observed physical descriptors available for comparison in the local metadata; semantic and stability scores are excluded from this physical-descriptor plot, and void fraction/porosity is excluded because void fraction is unavailable for all local records. Selected top- K values are averaged over the main saved single-run ranking artifact.

Missing-Data and Diversity Regression Tests

The final repository includes focused regression tests for missing descriptors, score clipping, full-precision distance calculation, symmetric masked distances, identical versus different active vectors, top- K pairwise-distance averaging, LEA candidate identifier preservation, and the exclusion of unavailable void fraction from numerical ranking. The test output is stored in `artifacts/diversity_revision/test_results.txt`; the completed local run reports 21 passing tests.

CO₂ Adsorption Scenario

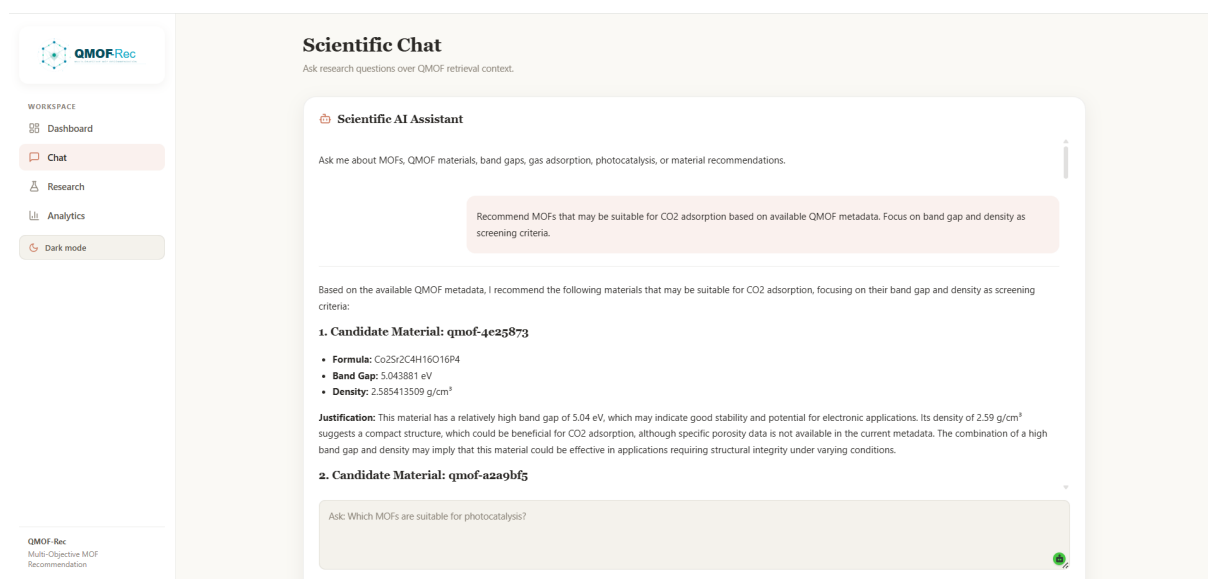


Figure S2: CO₂ adsorption scenario. The assistant recommends candidates based on available metadata and states that porosity and CO₂ uptake data are unavailable. Void fraction and adsorption uptake are unavailable in the current metadata.

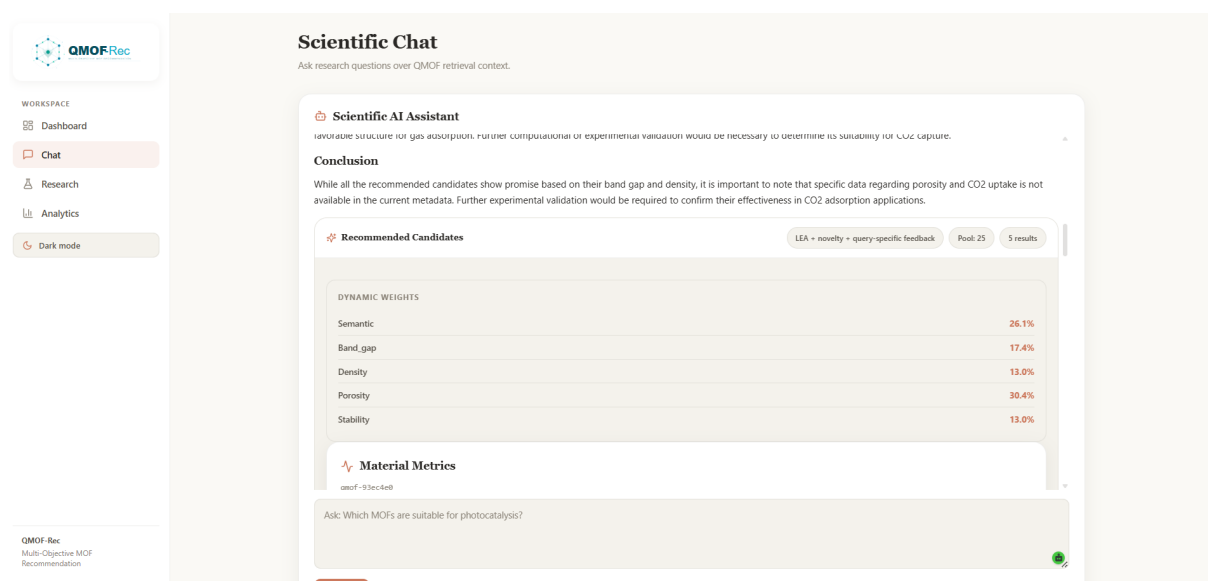


Figure S3: CO₂ adsorption scenario with structured candidate justifications and missing-descriptor warnings. Screenshots demonstrate the implemented interface and are not part of the offline ranking validation.

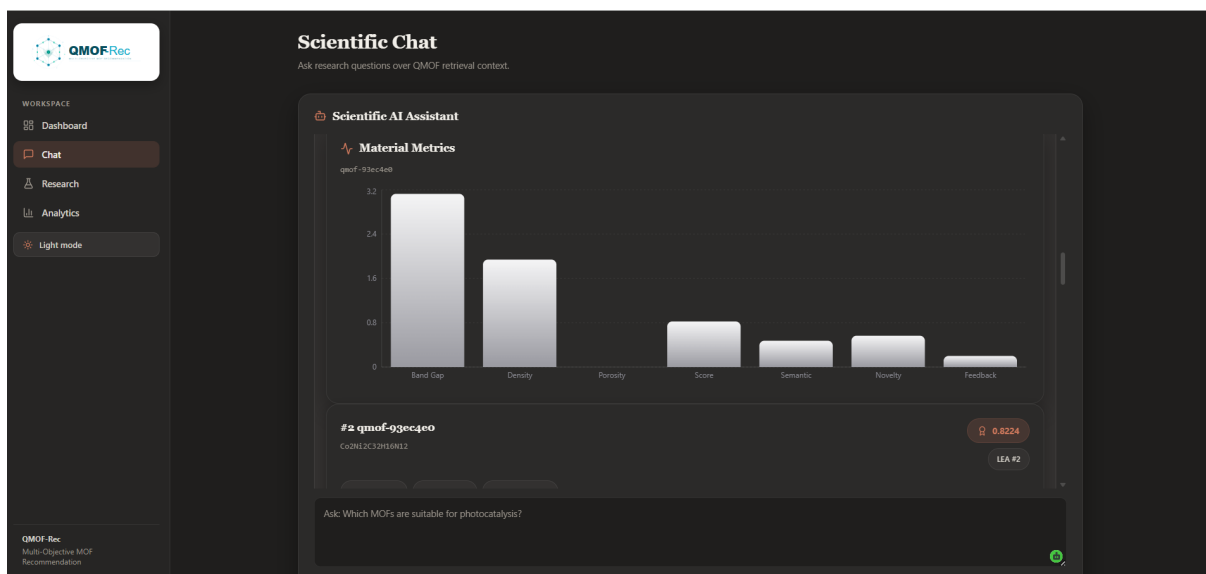


Figure S4: CO₂ adsorption scenario showing material metrics for a candidate record. In the revised implementation, void fraction is displayed only as unavailable metadata and is excluded from active numerical ranking.

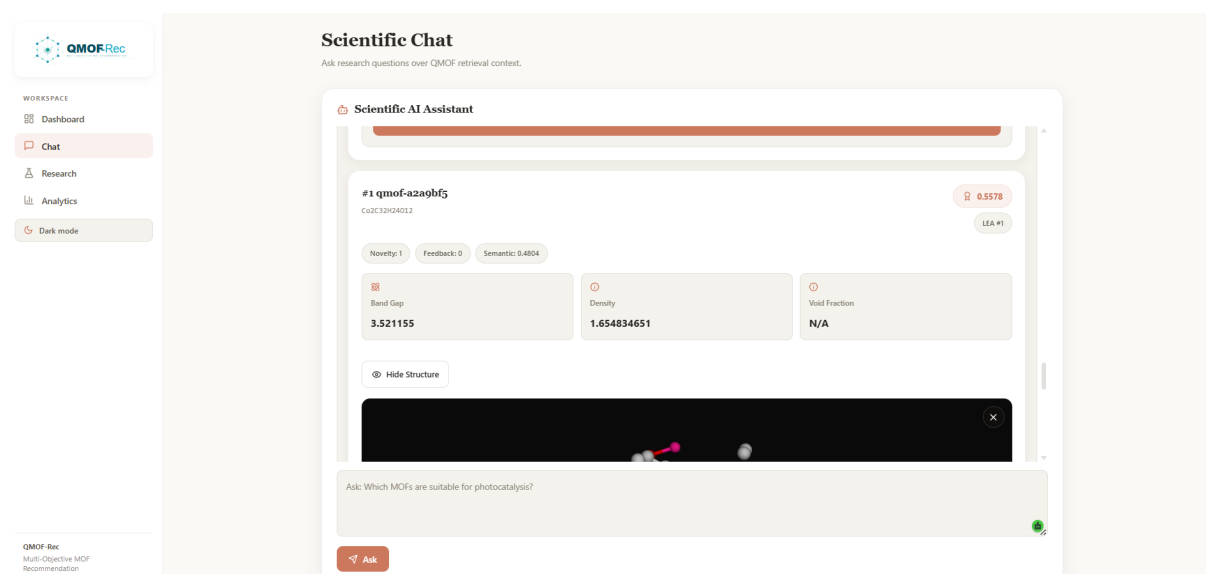


Figure S5: CO₂ adsorption scenario testing unsupported uptake claims. The assistant declines to identify experimentally validated CO₂ uptake because the retrieved context contains no experimentally validated uptake data.

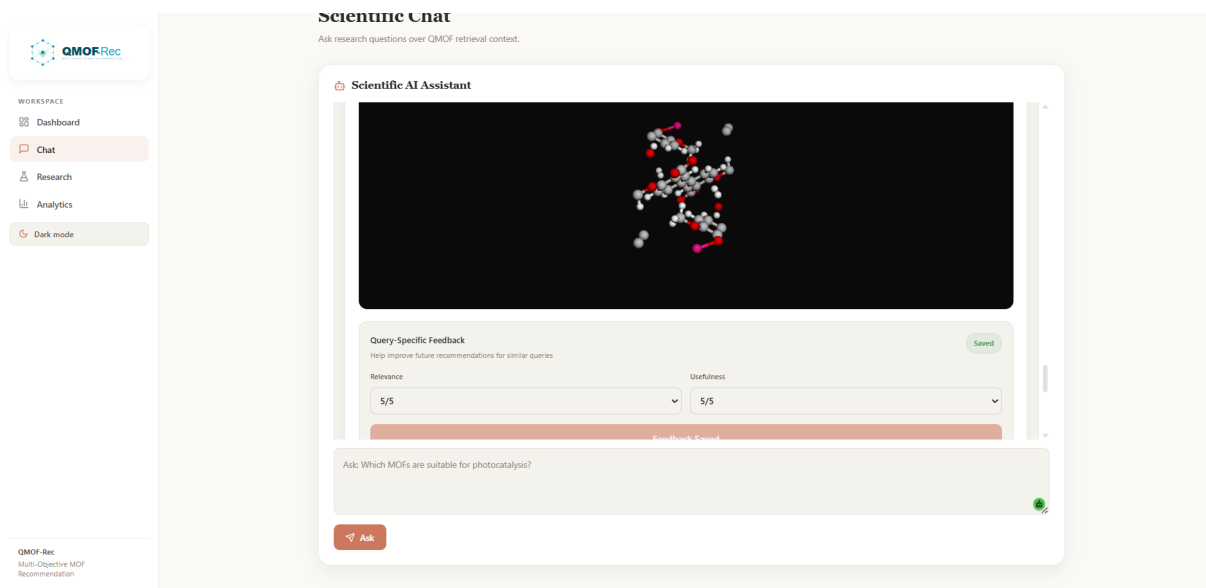
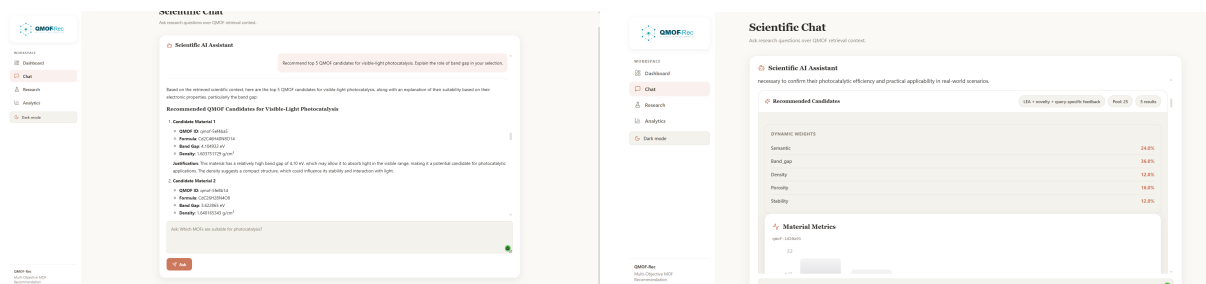


Figure S6: CO₂ adsorption scenario testing industrial-readiness claims. The assistant treats the listed MOFs as recommendation candidates requiring expert follow-up, not as experimentally validated capture materials.

Photocatalysis Scenario



(a) Top-5 photocatalysis candidates with band-gap-oriented justifications.

(b) Dynamic objective weights for a photocatalysis-oriented query.

Figure S7: Photocatalysis scenario screenshots. These illustrate answer structure and interface behavior rather than physical validation.

Missing-Information Trap



Figure S8: Missing-information trap screenshots. The assistant declines to fabricate experimental validation claims.

Lightweight Gas Storage

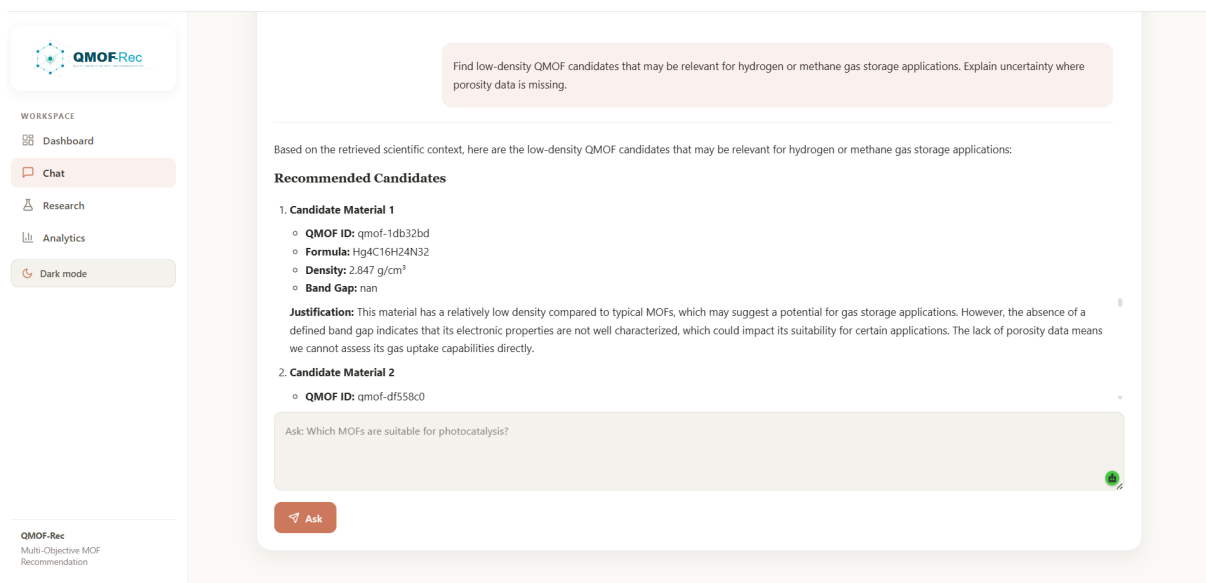


Figure S9: Lightweight gas-storage scenario. Candidate recommendations are based on available metadata such as density; void fraction, porosity, and gas uptake data are unavailable, so conclusions are preliminary.

Ranking Logic

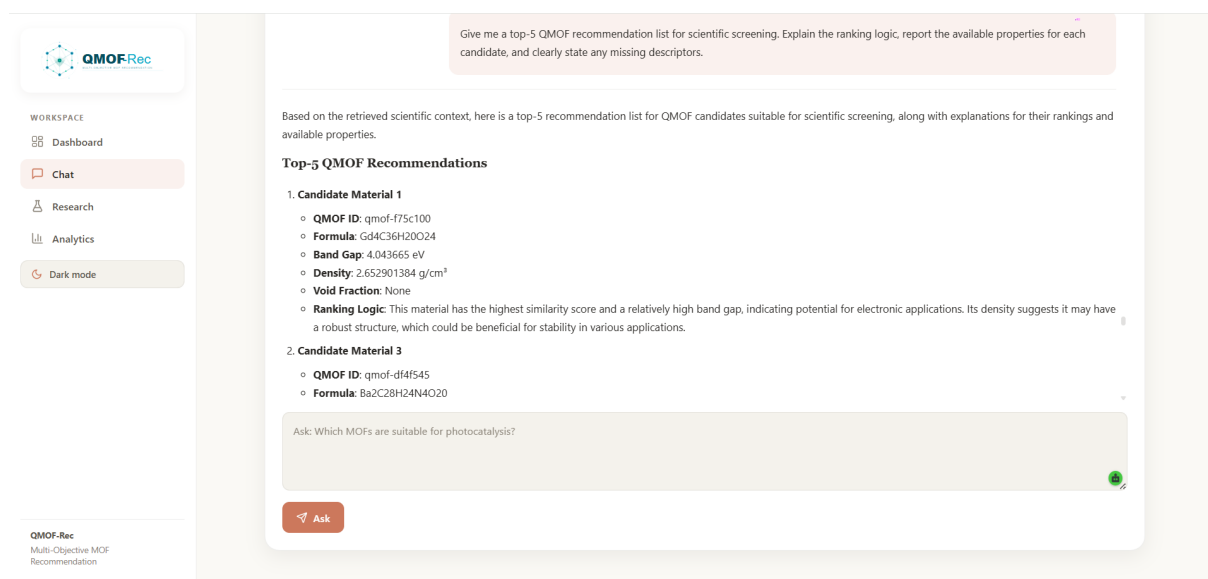


Figure S10: Ranking-logic scenario. The assistant provides a structured recommendation list with explicit ranking rationale and missing-descriptor reporting.

Chemically Novel Candidates

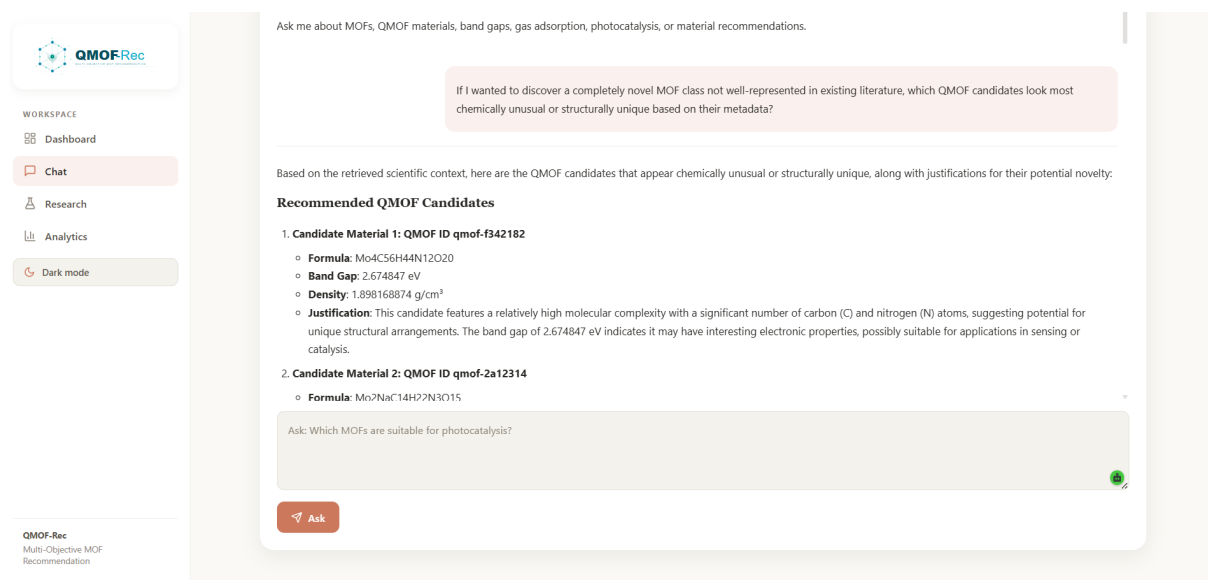


Figure S11: Chemically novel candidate scenario. The response is grounded in available meta-data and should not be interpreted as experimental, adsorption, porosity, or structure-level validation.